

Report - Task 1

1. Introduction

This report documents the work completed for Task 1 of the Redynox Cyber Security Internship: Introduction to Network Security Basics. The objective was to understand common network threats, implement basic security measures on a small network, monitor network traffic using Wireshark, and document findings.

2. Task Overview

2.1 Task 1: Introduction to Network Security Basics

Objective of the task:

- Understand different network threats (viruses, worms, trojans, phishing) and basic security concepts (firewalls, encryption, secure configurations).

Tools and methods used:

- Tools: Windows Defender Firewall, Router Web Interface, Nmap, Wireshark, Windows OS.
- Methods: Router hardening, firewall enablement, port scanning, packet capture and analysis.

Steps taken to complete the task:

1. Reviewed network security concepts: viruses, worms, trojans, phishing; firewalls; encryption; secure configs.
2. Logged into router web interface (usually 192.168.1.1) and changed default admin password.
3. Configured Wi-Fi to use WPA2 with AES on both 2.4GHz and 5GHz bands and disabled WPS.
4. Enabled Windows Defender Firewall on the host VM for host-level protection.
5. Performed network reconnaissance using Nmap to check open/filtered ports and services.
6. Installed Wireshark and captured network traffic on the active interface (DNS, HTTP, TLS).
7. Filtered and analyzed packets (dns, http, tls), inspected Client Hello and Server Hello messages, and saved captures.
8. Saved all evidence (screenshots and pcapng) and prepared this report.

Challenges faced and how they were overcome:

- Challenge: Capturing Wi-Fi monitor-mode traffic can be limited on Windows VMs due to adapter restrictions.

Resolution: Captured traffic on the host/bridged adapter and inside the VM where necessary. For TLS analysis, SNI in Client Hello was used to identify domains without needing to decrypt traffic.

- Challenge: Ensuring the captured packets included both unencrypted and encrypted traffic.

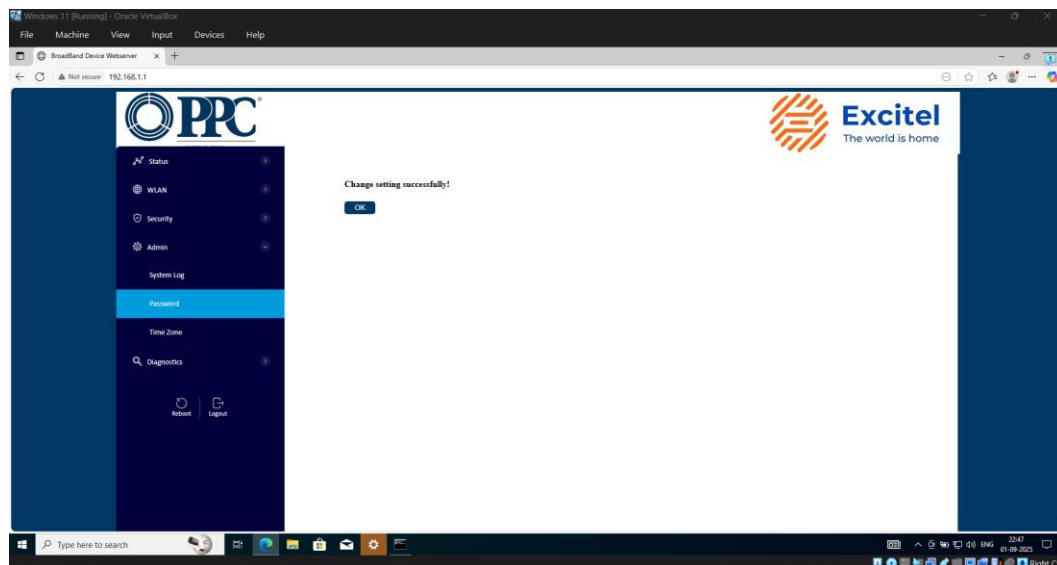
Resolution: Used a combination of visiting plain HTTP (neverssl.com) for HTTP captures and standard HTTPS sites for TLS handshakes.

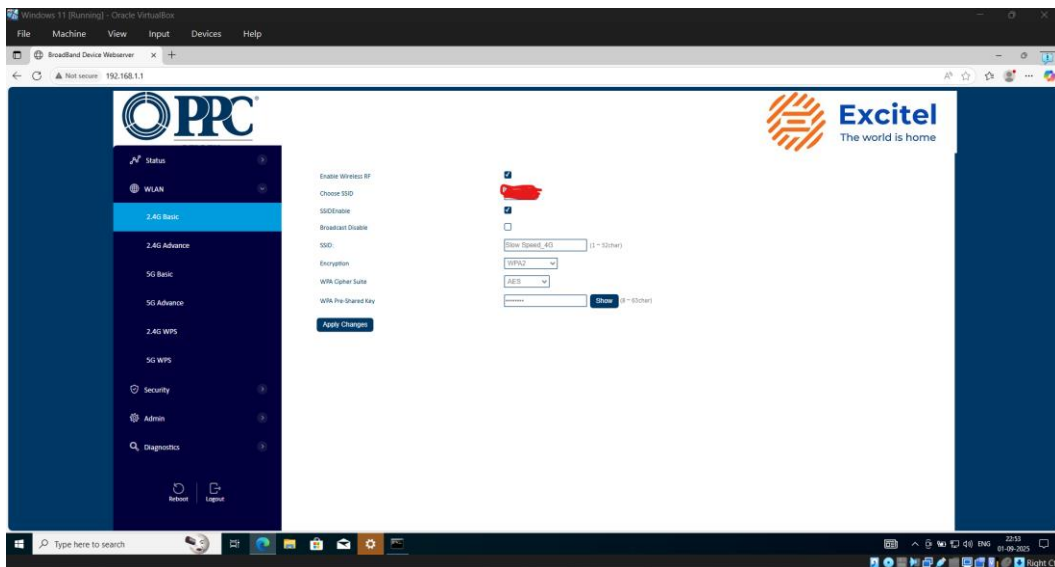
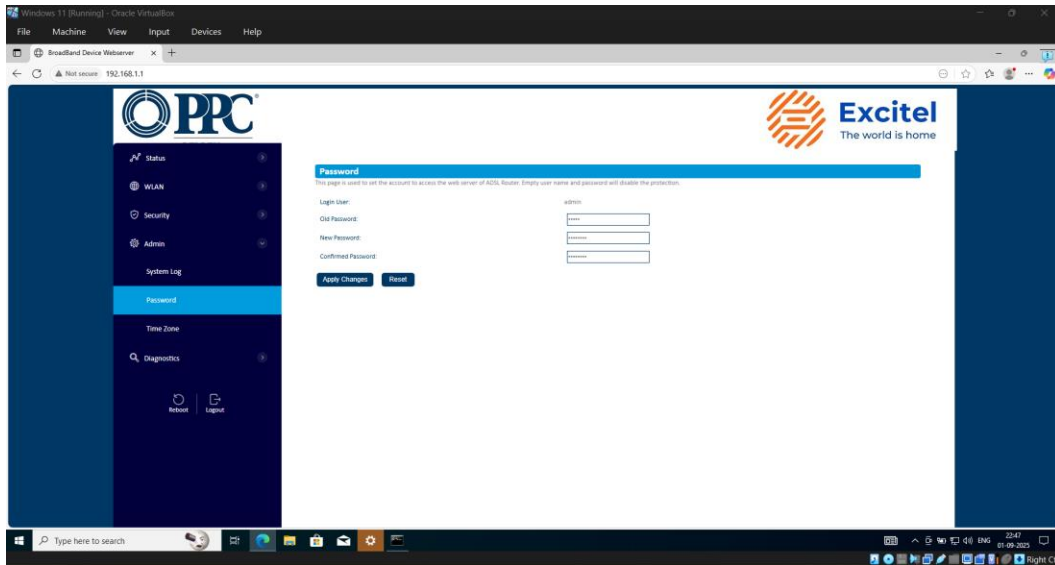
Results and outcomes:

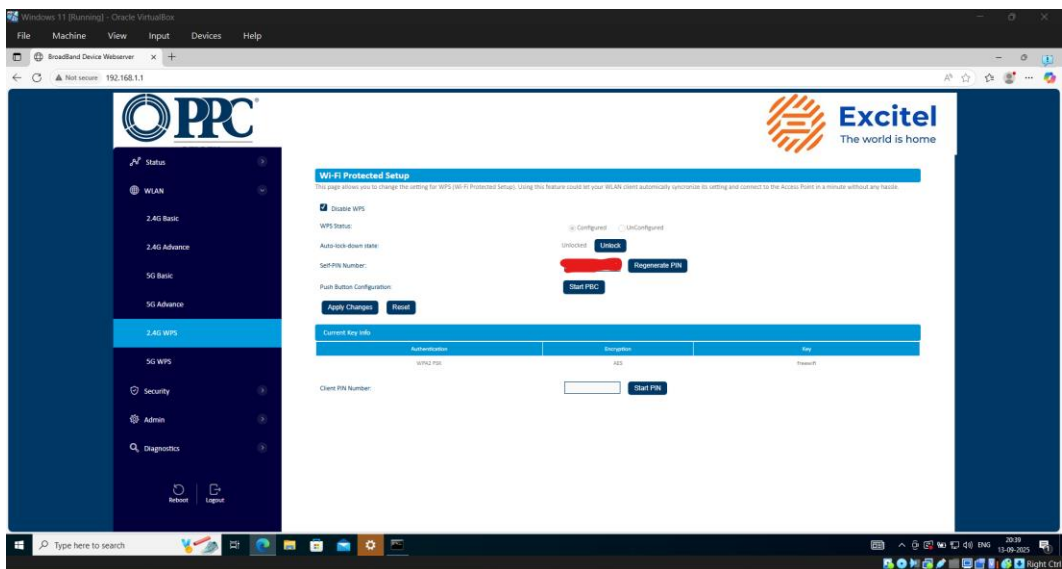
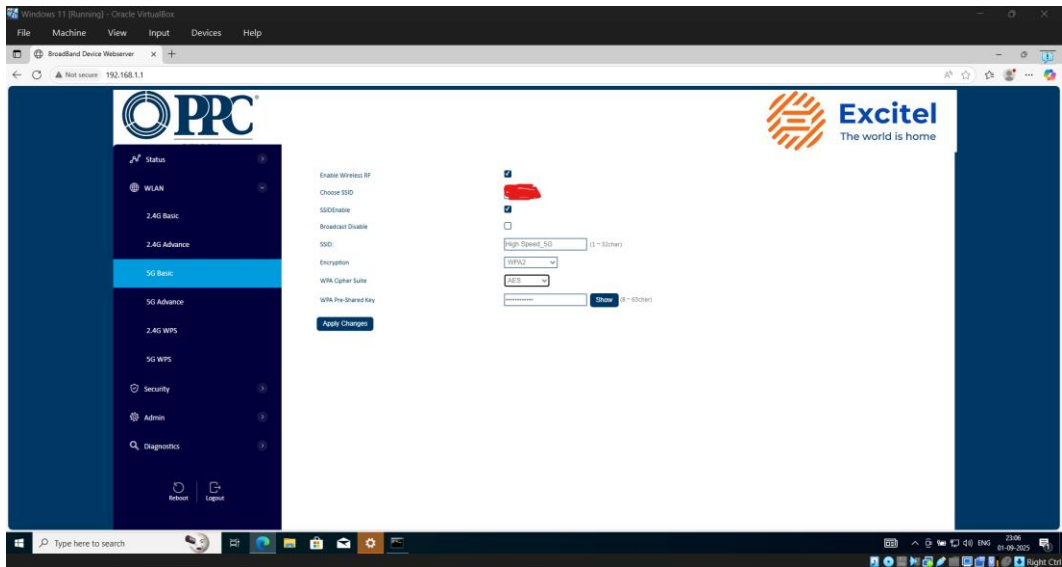
- Router configuration: default admin password changed, WPA2/AES enabled, WPS disabled (reduces attack surface).
- Firewall: Windows Defender Firewall enabled for host-level blocking of unauthorized inbound traffic.
- Nmap: Scans showed ports mostly filtered/closed, indicating limited exposure of services.
- Wireshark: Successful captures for DNS, HTTP (GET/200 OK), and TLS (Client Hello with SNI and Server Hello with selected cipher suite). This validated network monitoring capability and confirmed encrypted traffic for HTTPS.

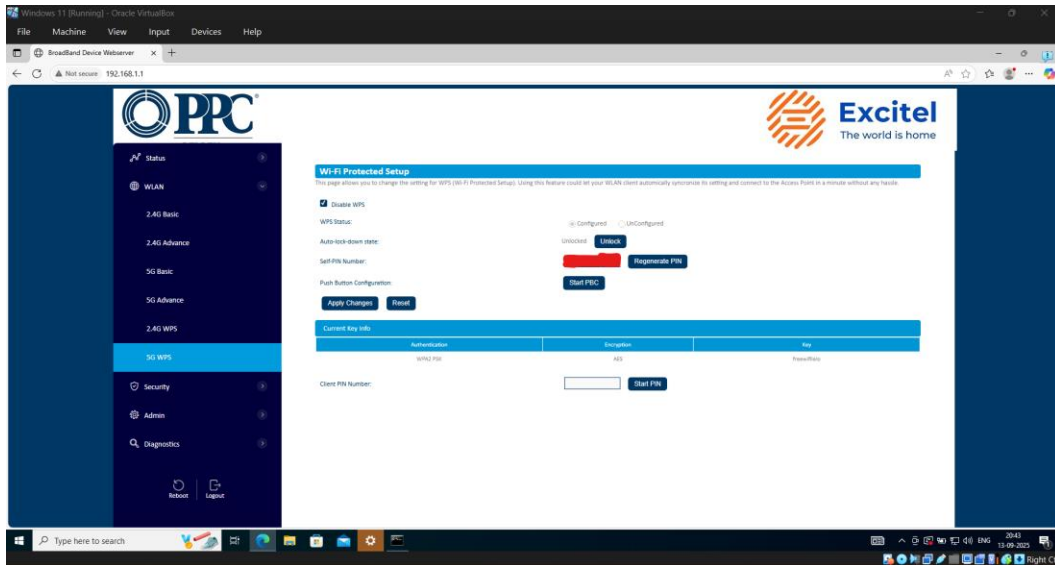
Evidence & Screenshots

Screenshot 2025-09-01 224728.png









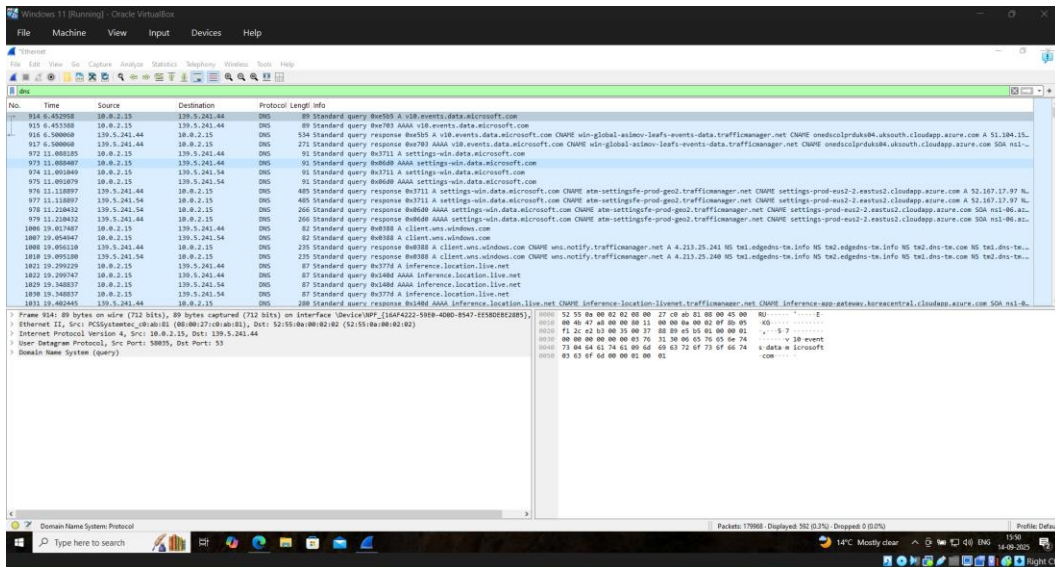
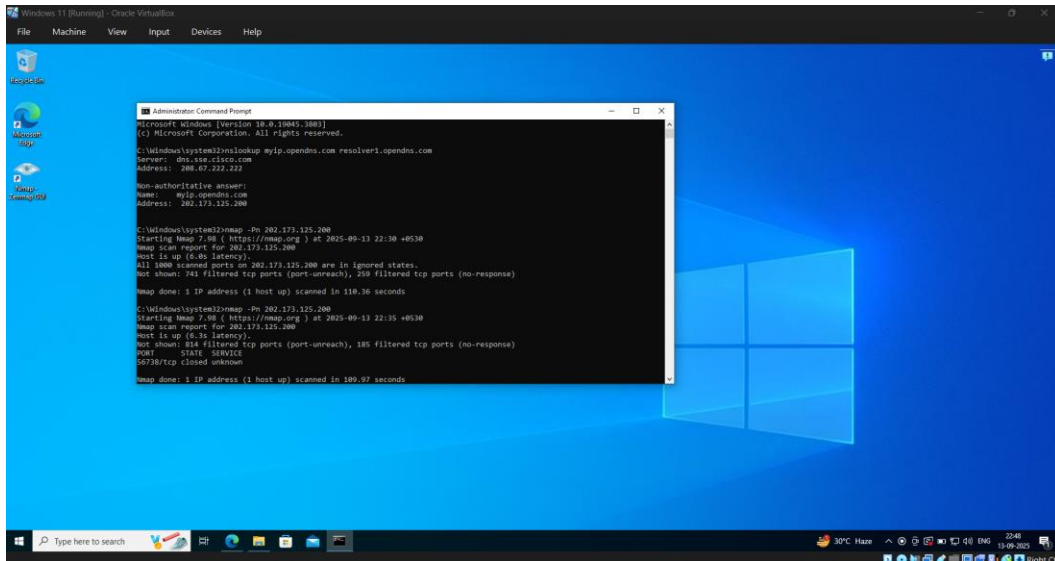
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Administrator: Command Prompt
Microsoft Windows [Version 10.0.19045.3803]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>nslookup myip.opendns.com resolver1.opendns.com
Server: dns.sse.cisco.com
Address: 208.67.222.222

Non-authoritative answer:
Name: myip.opendns.com
Address: 202.173.125.200

C:\Windows\system32>nmap -Pn 202.173.125.200
Starting Nmap 7.98 ( https://nmap.org ) at 2025-09-13 22:30 +0530
Nmap scan report for 202.173.125.200
Host is up (6.0s latency).
All 1000 scanned ports on 202.173.125.200 are in ignored states.
Not shown: 741 filtered tcp ports (port-unreach), 259 filtered tcp ports (no-response)
Nmap done: 1 IP address (1 host up) scanned in 110.36 seconds

C:\Windows\system32>nmap -Pn 202.173.125.200
Starting Nmap 7.98 ( https://nmap.org ) at 2025-09-13 22:35 +0530
Nmap scan report for 202.173.125.200
Host is up (6.3s latency).
Not shown: 814 filtered tcp ports (port-unreach), 185 filtered tcp ports (no-response)
PORT      STATE SERVICE
56738/tcp closed unknown
Nmap done: 1 IP address (1 host up) scanned in 109.97 seconds
```



Windows 11 (Running) - Oracle VM VirtualBox

File Machine View Input Devices Help

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Help

No.	Time	Source	Destination	Protocol	Length	Info
908	8.300319	93.123.17.254	10.0.2.15	HTTP	404	HTTP/1.1 200 Partial Content
3418	28.879318	10.0.2.15	93.123.17.254	HTTP	409	GET /filestreamservice/files/9499986-6a7f-426a-9f64-adb108dec1aa?fi=1757845548P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...
3424	29.800460	93.123.17.254	10.0.2.15	HTTP	842	HTTP/1.1 200 Partial Content
3415	29.801464	10.0.2.15	93.123.17.254	HTTP	474	GET /filestreamservice/files/9499986-6a7f-426a-9f64-adb108dec1aa?fi=1757845548P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...
3613	29.870207	93.123.17.254	10.0.2.15	HTTP	640	HTTP/1.1 200 Partial Content
4067	36.440242	10.0.2.15	34.196.13.28	HTTP	404	GET / HTTP/1.1
4053	36.773215	34.196.13.28	10.0.2.15	HTTP	1245	HTTP/1.1 200 OK (text/html)
4155	39.500453	10.0.2.15	34.196.13.28	HTTP	503	GET /?w=438&f=dc83d83d481254f8d6d1757845480.183.2.1.2&h=2612530308368-42=-336 HTTP/1.1
4108	40.292430	34.196.13.28	10.0.2.15	HTTP	940	HTTP/1.1 302 Found
4086	50.942566	10.0.2.15	93.123.116.253	HTTP	361	GET /filestreamservice/files/9482761-3958-4f05-5086-e5a6e40bf291?fi=1757845528P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...
4020	51.240864	93.123.116.253	10.0.2.15	HTTP	595	HTTP/1.1 200 OK
4001	51.762134	10.0.2.15	93.123.116.253	HTTP	361	GET /filestreamservice/files/9482761-3958-4f05-5086-e5a6e40bf291?fi=1757845528P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...
4013	52.801021	93.123.116.253	10.0.2.15	HTTP	345	HTTP/1.1 200 OK
4064	52.376380	10.0.2.15	93.123.17.254	HTTP	473	GET /filestreamservice/files/9482761-3958-4f05-5086-e5a6e40bf291?fi=1757845528P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...
4088	52.620055	93.123.17.254	10.0.2.15	HTTP	844	HTTP/1.1 200 Partial Content
4087	52.620055	93.123.17.254	10.0.2.15	HTTP	844	HTTP/1.1 200 Partial Content
4088	52.627096	10.0.2.15	93.123.17.254	HTTP	470	GET /filestreamservice/files/9482761-3958-4f05-5086-e5a6e40bf291?fi=1757845528P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...
4090	52.626362	10.0.2.15	93.123.17.254	HTTP	409	GET /filestreamservice/files/9482761-3958-4f05-5086-e5a6e40bf291?fi=1757845528P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...
5130	52.933059	10.0.2.15	93.123.17.254	HTTP	409	GET /filestreamservice/files/9482761-3958-4f05-5086-e5a6e40bf291?fi=1757845528P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...
5134	52.933088	10.0.2.15	93.123.17.254	HTTP	405	GET /filestreamservice/files/9482761-3958-4f05-5086-e5a6e40bf291?fi=1757845528P2-4040473-204-02763845?getQm6CvseJexCcoSHOfcLAVUMWLn3yCvU873bLIPU1YD=AKX28RyP92fsepe3tZf...

Frame 908: 420 bytes on wire (3360 bits), 420 bytes captured (3360 bits) on interface DeviceVirtIO (164f4222-5908-4080-8547-E53D0E8E2D28)

Ethernet II, Src: VSCSystem00:00:00:00:00:00, Dst: PCSystem00:00:00:00:00:00 (08:00:27:00:00:00)

Internet Protocol Version 4, Src: 93.123.17.254, Dst: 10.0.2.15

Transmission Control Protocol, Src Port: 80, Dst Port: 50274, Seq: 28000005, Ack: 1, Len: 372

[INFO] Reassembled TCP Segments (1049357 bytes): #1(1440), #2(1440), #3(1440), #4(15), #5(1440), #6(1440), #7(1440), #8(15), #9(15), #10(1440), #11(1440)

Hypertext Transfer Protocol

Data (1048576 bytes)

Frame (420 bytes): Reassembled TCP (1048576 bytes)

Hypertext Transfer Protocol

Packets: 17098 - Displayed: 987 (5.7%) - Dropped: 6 (0.0%)

Tempo to rise: 1551 14:09:2023

Right Ctrl

Windows 11 (Running) - Oracle VM VirtualBox

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File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Help

No.	Time	Source	Destination	Protocol	Length	Info
921	6.806296	10.0.2.15	51.184.15.253	TLSv1.2	268	Client Hello (501=0x.events.data.microsoft.com)
920	7.120774	51.184.15.253	10.0.2.15	TLSv1.2	1489	Server Hello, Certificate, Server Key Exchange, Server Hello Done
927	7.280215	10.0.2.15	51.184.15.253	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
920	7.487155	51.184.15.253	10.0.2.15	TLSv1.2	174	Change Cipher Spec, Encrypted Handshake Message, Application Data
930	7.584622	10.0.2.15	51.184.15.253	TLSv1.2	141	Application Data
931	7.586121	10.0.2.15	51.184.15.253	TLSv1.2	92	Application Data
932	7.586902	10.0.2.15	51.184.15.253	TLSv1.2	92	Application Data
933	7.588403	10.0.2.15	51.184.15.253	TLSv1.2	1479	Application Data
930	7.751104	51.184.15.253	10.0.2.15	TLSv1.2	92	Application Data
940	8.497389	51.184.15.253	10.0.2.15	TLSv1.2	339	Application Data
940	8.498484	51.184.15.253	10.0.2.15	TLSv1.2	96	Application Data
940	8.498484	10.0.2.15	51.184.15.253	TLSv1.2	268	Client Hello (501=0x.events.data.microsoft.com)
950	8.495434	10.0.2.15	51.184.15.253	TLSv1.2	1489	Server Hello, Certificate, Server Key Exchange, Server Hello Done
957	9.377084	51.184.15.253	10.0.2.15	TLSv1.2	212	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
950	9.376640	10.0.2.15	51.184.15.253	TLSv1.2	174	Change Cipher Spec, Encrypted Handshake Message, Application Data
959	9.386124	10.0.2.15	51.184.15.253	TLSv1.2	141	Application Data
960	9.386623	10.0.2.15	51.184.15.253	TLSv1.2	92	Application Data
961	9.386803	10.0.2.15	51.184.15.253	TLSv1.2	92	Application Data
965	9.426376	51.184.15.253	10.0.2.15	TLSv1.2	1450	Application Data
961	9.440000	51.184.15.253	10.0.2.15	TLSv1.2	339	Application Data

Frame 921: 268 bytes on wire (2144 bits), 268 bytes captured (2144 bits) on interface DeviceVirtIO (164f4222-5908-4080-8547-E53D0E8E2D28)

Ethernet II, Src: PCSystem00:00:00:00:00:00, Dst: 51:18:45:00:02:02 (52:55:0a:00:02:02)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 51.184.15.253

Transmission Control Protocol, Src Port: 50178, Dst Port: 443, Seq: 1, Ack: 1, Len: 214

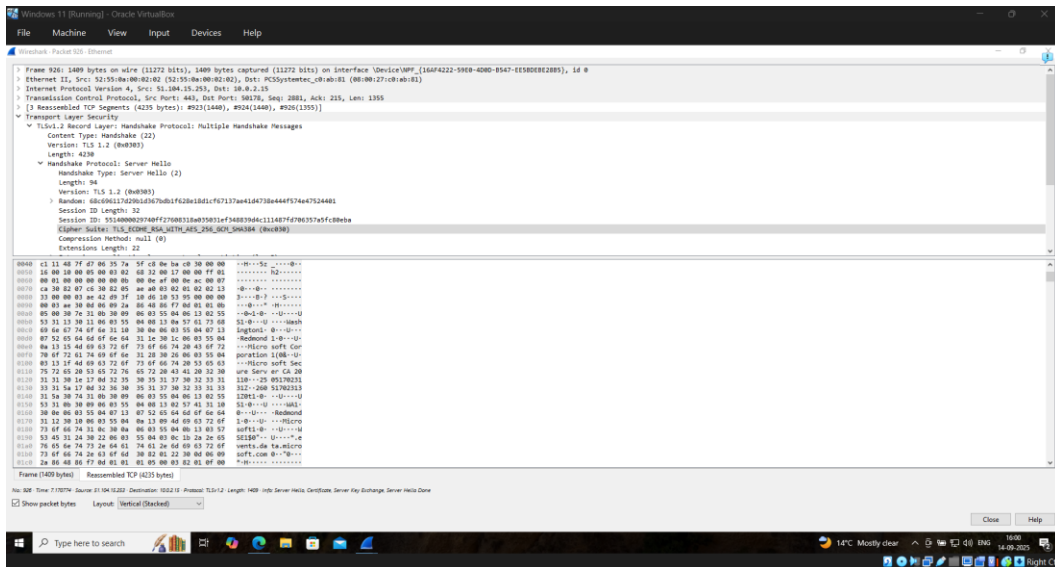
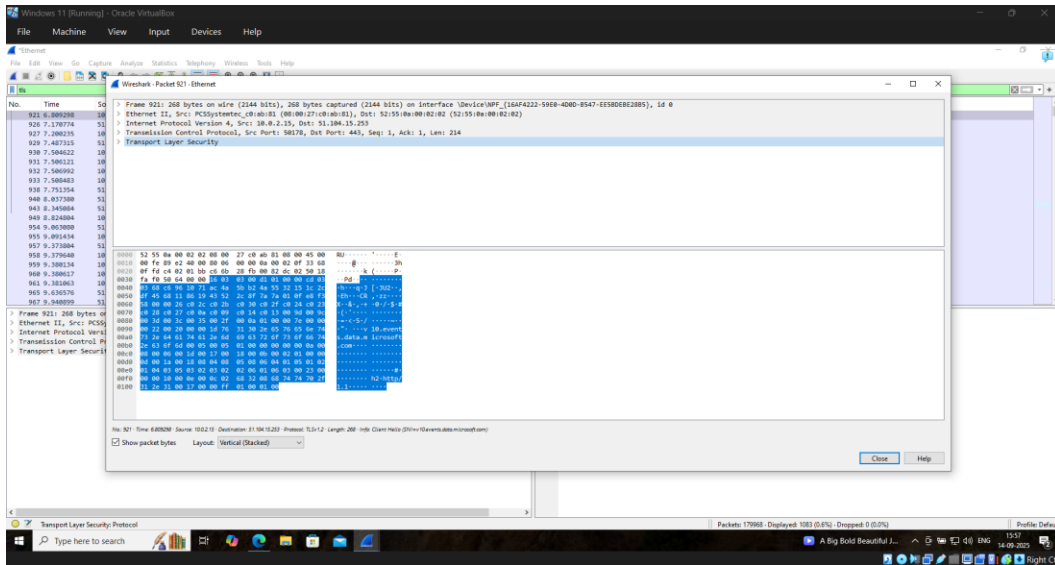
Transport Layer Security

Frame (268 bytes): Reassembled TCP (2144 bits)

Packets: 17098 - Displayed: 1083 (6.3%) - Dropped: 6 (0.0%)

Tempo to rise: 1551 14:09:2023

Right Ctrl



3. Conclusion

Task 1 successfully demonstrated the fundamentals of securing a small network and monitoring traffic. Implemented measures (router hardening, WPA2, disabling WPS, enabling firewall) reduced the risk of unauthorized access. Wireshark captures validated DNS resolution, HTTP requests, and TLS handshakes, confirming that sensitive communications use encryption. For larger networks, additional controls such as IDS/IPS, VPNs, centralized logging (SIEM), and segmentation would be recommended.

4. Closing Remarks

I acknowledge that all work performed relates to my learning objectives and adheres to the organization's confidentiality requirements. I appreciate the opportunity to learn practical network security fundamentals during this internship.